Sofpour®
heat pump

CENTRALISED WATER HEATING SOLUTION

Economical, safe, eco-friendly and hygienic hot water

CERTIFIED ISO 9001:2015 COMPANY

75%* ENERGY SAVING

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CENTRALISED WATER HEATING SOLUTION

Sofpour Thermo Technology Division has been providing hot water solutions for more than a decade. Sofpour brings you Air Source Heat Pump Water Heaters for all your hot water requirements. Sofpour Heat Pumps are available in a variety of combinations ranging from 200 Liters to1000 Liters systems for domestic use. Higher capacity systems are also available for commercial and industrial use.

Sofpour Heat Pump Water Heaters are easy to install and can be placed in a bathroom, terrace or rooftop of a building. It is a compact centralised heating solution that can be installed without compromising the aesthetics of the building.

Major Utility Sector

Residential Bungalow

Residential Apartment

Hospital & Clinic

Hotel & Restaurant

Airport & Station

Industrial & Commercial

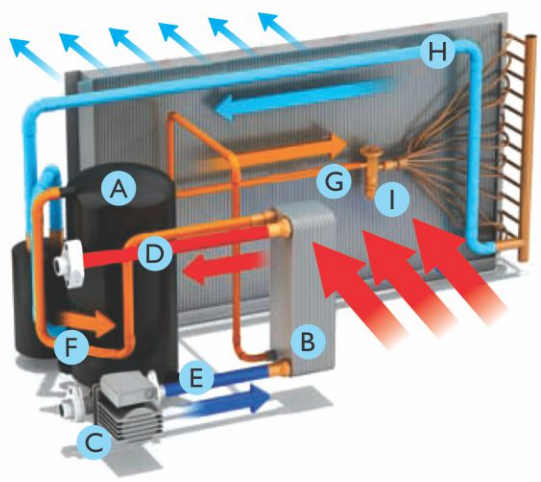
Hostel & Guest House

Health Spa & Gymnasium

Sofpour Heat Pumps are designed to deliver hot water up to 55°C in any weather condition. In a Heat Pump electricity is used to run the compressor and not to heat the water, therefore it is more economical and saves 75% electricity as compared to conventional electric water heaters.

Sofpour Heat Pump - Working Principle

- Hot air (at room temperature) is blown inside the heat pump by a fan and flows through the evaporator.
- Liquid refrigerant in the evaporator absorbs heat from the air and evaporates.
- Gas refrigerant now enters the compressor and is compressed towards higher pressure state thereby increasing its temperature.
- Gas refrigerant at high temperature and pressure enters the condenser and gets cooled which releases massive heat into the heat exchanger.
- This heat exchanger transfers heat to water and this hot water is transferred to storage tank by a pump.
- The hot water pump circulates water between storage tank and heat exchanger.
- This is called "Circle Heating" in case of split type heat pump.
- Liquid refrigerant flows through an expansion valve where pressure and temperature further decreases to build up evaporating condition in the evaporator to be ready for the next step.
- This thermodynamic cycle is repeated until desired temperature of hot water is achieved in the storage tank.



Circuit Description

A. Compressor

B. Heat Exchanger

C. Re-circulation Pump

D. Hot Water Outlet to Tank

E. Water Inlet from Tank

F. Hot Refrigerant

G. Cooled Liquid Refrigerant

H. Evaporated Refrigerant Vapour

I. Expansion Valve

Commercial Unit

Storage Tank

Domestic Unit

Key Features

- Weather Resistant Body
- Easy to Install & Service
- Enamel Tank for Long Life
- Environment Friendly Technology
- Better COP and Pocket Friendly
- 5 Bar Working Pressure
- Hygienic and Clean Water
- Auto Defrost Works in Low Temperatures
- Isolated Safe Condenser
- High Efficiency Compressor
- Magnesium Anode
- Quick Heating
- Intelligent Controller
- Easy & Smooth Functioning
- Energy Efficient
- Power Saving

Key Highlights

- Working condition range: -15°C to 45°C.
- Constant Hot Water Output at 55°C (Domestic Heat Pump) irrespective of weather conditions. Higher temperature can be attained for industrial usage.
- Environment friendly R410a refrigerant.
- Highly efficient Shell and Tube heat exchanger for higher COP and lower operating cost.
- Inbuilt High Pressure Protection System for enhanced safety.
- Panasonic or Copeland Scroll Compressor.
- Automatic Defrost Function.
- Smart System Controller (Circuit, LCD Panel, Temp. Sensors).

Key Applications

- Shower & Bathroom
- Hand Washing
- Dish Washing
- Cooking & Baking
- Washing Clothes
- Swimming Pool
- Jacuzzi & Spa

Sofpour Commercial Air Source Heat Pump Water Heater

Sofpour Commercial Heat Pumps can provide hot water round the clock and throughout the year in an energy efficient way. They are ideal for commercial use in Hotels, Resorts, Hospitals, Spas, Swimming Pools and Residential Apartments. Installing heat pump can save huge running and maintenance cost. Once installed they do not need any human interference for daily operations. Hot water can be ensured every single time without getting worried of the procedure of the system.

Heat Pumps have a COP (Coefficient of Performance). It is an important energy efficiency index of Heat Pump Water Heater.

$$COP = \frac{\text{Output Energy of Heat Pump Water Heater}}{\text{Input Energy of Heat Pump Water Heater}}$$

Mostly, Heat Pump Water Heater COP is greater than > 4.

Technical Specifications

SN	Model	SHP-200 E	SHP-200 H	SHP-300 E	SHP-300 H	SHP-300 X	SHP-500 E	SHP-500 H	SHP-500 X	SHP-1000 E	SHP-1000 H
1	Capacity (Liters)	200	200	300	300	300	500	500	500	1000	1000
2	Product Type	Split Type									
3	Power Input (kW)	0.95	1.4	0.95	1.4	1.85	1.4	1.85	2.6	2.6	4.7
4	Power Output (kW)	4	6	4	6	7.9	6	7.9	11	11	20.4
5	Hot Water Output L/hr	86	129	86	129	170	129	170	237	237	439
6	Heating Time*	2 hrs 19 min	1 hrs 33 min	3 hrs 30 min	2 hrs 19 min	1 hrs 45 min	3 hrs 52 min	2 hrs 56 min	2 hrs 06 min	4 hrs 13 min	2 hrs 16 min
7	Max. Hot Water Temp. HP Mode (°C)	55									
8	Max. Hot Water Temp. with Electrical Heating Element (°C)	75									
9	Noise Level (dB)	< 48	< 48	< 48	< 48	< 51	< 51	< 51	< 54	< 54	< 58
10	Refrigerant	R410a									
11	Net Weight of Outdoor Unit (kg)	56	65	56	65	70	65	70	85	85	160
12	Dimension (LWH) (in mm)	965 365 560	965 365 560	965 365 560	965 365 560	1100 410 600	965 365 560	1100 410 600	1120 410 725	1120 410 725	750 750 1080
13	Power Supply V/Ph/Hz	220-240 / 1 / 50									
14	Water Circulation Pump	Built - in									
15	Tank Type & MOC	Vertical / Horizontal - Glass Lined / HDGI / SS304 L									
16	Operating Pressure (bar)	5									
17	Testing Pressure (bar)	8									
18	Heating Element (kW)	2		3						6	
	Testing Conditions	Inlet Water 15°C / Outlet Water 55°C Dry bulb / Wet bulb temperature 20°C / 15°C *Values are approximate & subject to our lab conditions									

Note : Pictures shown in the brochure are for reference only & actual product may vary • Advisable to install the unit with voltage stabilizer as per site conditions • To the best of our knowledge the information contained in this publication is accurate • Sofpour Innovations Pvt Ltd. maintains a policy of continuous development and hence given technical specifications are subject to change without prior notice

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SCAN THIS TO KNOW MORE

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